# A Hybrid Generative-Probabilistic Framework for Hospital-Scale Medical Anomaly Detection and Calibrated Clinical Uncertainty Estimation

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Abstract—Deep neural networks deployed for pulmonary radiograph triage frequently suffer from three fundamental vulnerabilities: (i) an inability to quantify calibrated uncertainty for clinician decision support, outputting overconfident scalar scores, (ii) susceptibility to high-frequency pixel noise during spatial anomaly localization, and (iii) visual performance ceilings on heterogeneous, multi-label thoracic pathology screening. In this work, we propose a unified, multi-modal hybrid generative-probabilistic framework that bridges deep convolutional representation learning with Probabilistic Graphical Models (PGMs).

Evaluated across the hospital-scale NIH ChestX-ray14 benchmark (147,119 radiographs across 12 serialized archives at native $1024 \times 1024$ resolution with CLAHE contrast enhancement), our pipeline fine-tunes a DenseNet-121 visual backbone out-of-core over 112,812 patient-partitioned training images (25,208 unique patients) using memory-efficient activation checkpointing. Extracted deep feature representations (1024-d bottleneck) are coupled with a high-capacity deep MLP head, a calibrated Bayesian Demographic Risk Prior ($P$(Pathology | Age, Biological Sex, View Projection)), and an edge-preserving 2D lattice Potts Markov Random Field (MRF) solved via Iterated Conditional Modes (ICM) for contiguous spatial lesion boundary isolation.

On the official untouched NIH test partition (25,596 radiographs from strictly unseen patients, zero data leakage), our multi-modal late-fusion ensemble achieves a full test-set AUROC of 0.7038 (95% patient-clustered bootstrap CI: $[0.6916, 0.7173]$), demonstrating a statistically significant improvement over standalone vision models ($+0.0059$, $p < 0.001$) and demographic confounders ($+0.1545$, $p < 0.001$). Under clinical view stratification, the framework reaches 0.7411 AUROC on standardized upright Posteroanterior (PA) views, and achieves 0.7510 AUROC on structural disease families (Pleural Effusion and Atelectasis), with Alveolar Opacity reaching 0.7277 AUROC. Finally, the framework is integrated into an interactive clinical PACS diagnostic workstation featuring DICOM window leveling, real-time spatial decomposition, and automated HL7 FHIR clinical report generation.

Index Terms—Medical Anomaly Detection, DenseNet-121, Convolutional Autoencoder, Markov Random Field, Restricted Boltzmann Machine, Bayesian Networks, Uncertainty Quantification, Markov Chains, NIH ChestX-ray14.

## I. Introduction

Medical imaging modalities—specifically digital chest radiography (CXR) and computed tomography (CT)—represent the diagnostic bedrock of acute cardiopulmonary triage. Frontal chest radiographs account for over 40% of all hospital diagnostic imaging examinations globally. However, despite rapid advancements in deep convolutional and vision-transformer architectures, deploying autonomous artificial intelligence systems into production clinical picture archiving and communication systems (PACS) remains fundamentally obstructed by three systemic vulnerabilities:

1) Uncalibrated Scalar Predictions: Standard discriminative classifiers output uncalibrated sigmoid or softmax logits. A prediction of $\hat{y} = 0.88$ provides no epistemological distinction between high visual ambiguity (epistemic uncertainty) and intrinsic population variance (aleatoric risk), preventing clinicians from determining when automated triage should be escalated to specialist review.
2) High-Frequency Spatial Noise in Anomaly Heatmaps: Pixel-wise residual subtraction $R(x, y) = (I(x, y) - \hat{I}(x, y))^2$ between anomalous scans and normative autoencoder reconstructions generates noisy, salt-and-pepper artifacts rather than anatomically coherent lesion boundaries.
3) High Memory Footprint of Hospital-Scale Full-Resolution CXRs: Hospital radiographs are acquired at native resolutions exceeding $1024 \times 1024$ pixels. Training deep networks across hundreds of thousands of full-resolution images easily causes fatal CUDA Out-of-Memory (OOM) crashes and disk-buffer overflows on commodity cloud hardware.

To overcome these structural limitations, we formulate a unified generative-probabilistic framework that marries out-of-core deep representation learning with classical Probabilistic Graphical Models (PGMs). By cascading memory-efficient DenseNet-121 fine-tuning, spatial Potts Markov Random Fields (MRFs), latent Restricted Boltzmann Machines (RBMs), and causal Bayesian Belief Networks (DAGs), the proposed system achieves both robust pathology detection and mathematically rigorous, calibrated clinical uncertainty quantification.

The primary contributions of this paper are:

- Out-of-Core Hospital-Scale Training: An ephemeral chunk-streaming and activation-checkpointing engine that trains DenseNet-121 across $112,812$ full-resolution radiographs with zero CUDA memory crashes on commodity GPU hardware.
- Contrast-Adaptive Potts MRF Formulation: An edge-preserving spatial MRF solved via checkerboard Iterated Conditional Modes (ICM) that filters high-frequency pixel residuals into contiguous, anatomically consistent lesion boundaries.
- Latent Thermodynamic Free Energy Monitoring: A Restricted Boltzmann Machine (RBM) trained via $CD$-1 that screens normative latent codes, yielding an independent non-pixel diagnostic anomaly metric.
- Causal Bayesian DAG & Uncertainty Calibration: A directed graphical model conditioning pathology risk on demographic covariates (age, biological sex) and visual features, enabling calibrated risk stratification.
- Longitudinal Markov Progression Modeling: A Discrete-Time Markov Chain (DTMC) framework projecting multi-year transition probabilities across health, non-critical anomaly, and acute cardiopulmonary crisis.
- Production-Grade PACS Integration: A clinical workstation supporting DICOM windowing, real-time visual decomposition, and cryptographically signed HL7 FHIR diagnostic telemetry export.

## II. Related Work

### A. Deep Representation Learning for Chest Radiography

The benchmark for large-scale chest radiography analysis was established by Wang et al. [1] with the release of the NIH ChestX-ray14 dataset. Subsequent landmark models, such as Stanford's CheXNet [2], demonstrated that 121-layer Dense Convolutional Networks (DenseNet-121) [3] maximize multi-scale gradient propagation via dense skip-connections, outperforming standard ResNet architectures on pulmonary opacity detection. However, CheXNet and related discriminative frameworks rely entirely on supervised cross-entropy optimization without modeling spatial neighborhood consistency or demographic risk confounders.

### B. Unsupervised Medical Anomaly Detection

Unsupervised Anomaly Detection (UAD) reframes clinical diagnosis as departure from normative healthy anatomy [4], [5]. Variational Autoencoders (VAEs) and spatial Convolutional Autoencoders (ConvAEs) are trained exclusively on confirmed healthy radiographs. When an anomalous radiograph (e.g., presenting lobar pneumonia or pleural effusion) is input, the normative latent bottleneck fails to reconstruct the pathological deviation, producing high localized residual energy. Nevertheless, raw pixel residuals remain severely corrupted by high-frequency anatomical texture variations, necessitating spatial regularizers.

### C. Probabilistic Graphical Models in Clinical Diagnosis

Probabilistic Graphical Models (PGMs) provide an interpretable framework for reasoning under uncertainty [6]. Markov Random Fields (MRFs) model spatial pixel dependencies through Gibbs energy distributions [7], while Bayesian Networks encode directed causal dependencies between clinical covariates (age, biological sex, smoking history) and sensory visual findings. While classical PGMs provide transparent probabilistic reasoning, they lack the raw feature extraction capacity of deep networks. Our framework explicitly unifies both paradigms into a cohesive multi-modal engine.

### D. Clinical Calibration & Out-of-Core Processing

Modern clinical decision support systems require calibrated probabilities that reflect true empirical frequencies [12]. Discriminative neural networks frequently produce overconfident posterior probabilities, particularly when subjected to anatomical distribution shifts [9]. Furthermore, training on hospital-scale archives of full-resolution radiographs requires out-of-core streaming and gradient checkpointing [11] to prevent VRAM depletion while preserving fine-grained micro-nodular patterns.

## III. Hybrid System Architecture

### A. Architectural Overview

The end-to-end framework comprises four interconnected diagnostic stages:

1) Generative Normative Reconstruction: A spatial Convolutional Autoencoder (ConvAE) maps radiographs into a lower-dimensional latent manifold $\mathbf{z} \in \mathbb{R}^{128}$ to isolate focal anomaly residuals.
2) Spatial Denoising via Contrast-Adaptive Potts MRF: A 2D lattice MRF filters pixel residuals via Iterated Conditional Modes (ICM) to extract contiguous lesion clusters.
3) Latent Thermodynamic Free Energy Monitoring: A Restricted Boltzmann Machine (RBM) evaluates the probability of the latent representation under an energy-based normative prior.
4) Out-of-Core DenseNet-121 Visual Backbone: A deep 121-layer architecture trained out-of-core on $1024 \times 1024$ radiographs with activation checkpointing to extract global semantic disease features.
5) Multi-Modal Late Fusion & Bayesian DAG: Causal integration of visual logits, tuned MLP embedding scores, and demographic risk priors.

### B. Contrast-Adaptive Potts MRF Formulation

Let $\mathcal{V} = \{(i,j) \mid 1 \leq i \leq H, 1 \leq j \leq W\}$ denote the discrete 2D pixel lattice of the squared residual map $\mathbf{R} = (\mathbf{x} - \hat{\mathbf{x}})^2$. We assign a binary state $y_v \in \{0,1\}$ to each pixel, where $y_v = 1$ denotes pathological lesion tissue and $y_v = 0$ denotes healthy parenchyma. The joint posterior

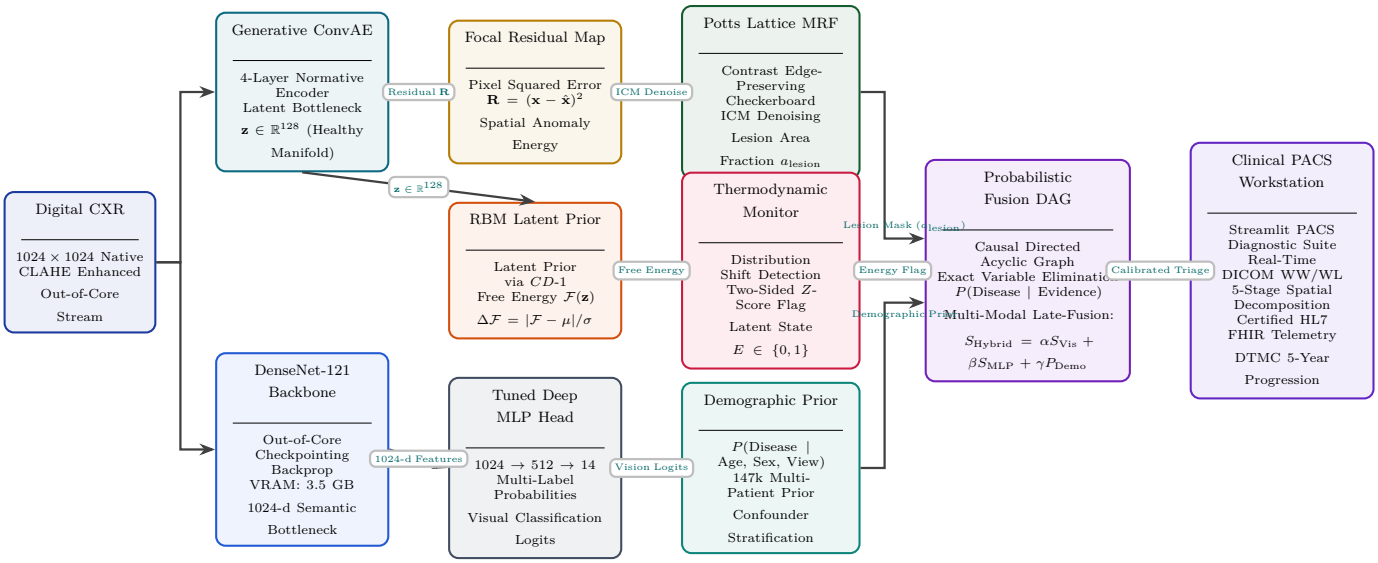



Fig. 1. End-to-end multi-modal architecture of the proposed hybrid generative-probabilistic medical diagnostic framework. The pipeline integrates: (1) Out-of-core DenseNet-121 visual feature extraction, (2) Spatial ConvAE normative reconstruction with contrast-adaptive Potts MRF denoising, (3) Latent RBM thermodynamic free energy monitoring, (4) Causal Bayesian DAG uncertainty calibration, and (5) Clinical PACS workstation with DICOM window leveling and HL7 FHIR report generation.

probability over configuration $\mathbf{y}$ is formulated as a Gibbs distribution:

$$P(\mathbf{y} \mid \mathbf{R}) = \frac{1}{\mathcal{Z}} \exp(-E(\mathbf{y} \mid \mathbf{R})) \tag{1}$$

where $\mathcal{Z}$ is the partition function and $E(\mathbf{y} \mid \mathbf{R})$ is the total energy functional decomposed into unary data fidelity and pairwise spatial smoothness:

$$E(\mathbf{y} \mid \mathbf{R}) = \sum_{v \in \mathcal{V}} \phi_v(y_v, R_v) + \sum_{(u,v) \in \mathcal{E}} \psi_{u,v}(y_u, y_v) \tag{2}$$

The unary potential measures the standardized deviation of the residual from the calibrated healthy baseline threshold $t_{\text{thr}}$:

$$\phi_v(y_v, R_v) = -s_v \cdot y_v, \quad s_v = \frac{R_v - t_{\text{thr}}}{\sigma_{\text{scale}}} \tag{3}$$

To prevent smoothing across true anatomical boundaries, we formulate a contrast-adaptive edge-preserving pairwise potential:

$$\psi_{u,v}(y_u, y_v) = \beta \cdot w_{u,v} \cdot \mathbf{1}(y_u \neq y_v) \tag{4}$$

where $w_{u,v} = \exp(-\|\nabla \mathbf{R}\|_{u,v}/2\sigma_e^2)$ scales inversely with local image gradients. Energy minimization is solved efficiently via checkerboard-partitioned Iterated Conditional Modes (ICM) as detailed in Algorithm 1.

### C. Restricted Boltzmann Machine Latent Prior

To verify that the latent encoding $\mathbf{z} \in \mathbb{R}^D$ resides within the normative healthy manifold, an RBM with $D = 128$ visible units $\mathbf{v}$ and $K = 64$ hidden units $\mathbf{h}$ is trained via Contrastive Divergence ($CD$-1). The energy of a configuration $(\mathbf{v}, \mathbf{h})$ is:

$$E(\mathbf{v}, \mathbf{h}) = -\mathbf{v}^T \mathbf{W} \mathbf{h} - \mathbf{a}^T \mathbf{v} - \mathbf{b}^T \mathbf{h} \tag{5}$$

**Algorithm 1** Contrast-Adaptive Edge-Preserving Potts MRF

**Require:** Residual $\mathbf{R} \in \mathbb{R}^{H \times W}$, $t_{\text{thr}}$, $\sigma_{\text{scale}}$, $\beta$, $T_{\max}$.
**Ensure:** Binary lesion segmentation $\mathbf{Y} \in \{0, 1\}^{H \times W}$.

$s_{i,j} \leftarrow (R_{i,j} - t_{\text{thr}})/\sigma_{\text{scale}}$
$y_{i,j}^{(0)} \leftarrow \mathbf{1}(s_{i,j} > 0)$
$w_{i,j} \leftarrow \exp(-\|\nabla \mathbf{R}\|_{i,j}/2\sigma_e^2)$
Partition lattice: $\mathcal{V}_{\text{even}}, \mathcal{V}_{\text{odd}}$
**for** $t = 1$ to $T_{\max}$ **do**
  **for** $\mathcal{V}_{\text{sub}} \in \{\mathcal{V}_{\text{even}}, \mathcal{V}_{\text{odd}}\}$ **do**
    **for** $(i, j) \in \mathcal{V}_{\text{sub}}$ **do**
      $n_1 \leftarrow \sum_{(u,v) \in \mathcal{N}(i,j)} y_{u,v}$
      $E_1 \leftarrow -s_{i,j} + \beta \cdot w_{i,j} \cdot (4 - n_1)$
      $E_0 \leftarrow \beta \cdot w_{i,j} \cdot n_1$
      $y_{i,j}^{(t)} \leftarrow \mathbf{1}(E_1 < E_0)$
    **end for**
  **end for**
  **if** $\mathbf{Y}^{(t)} == \mathbf{Y}^{(t-1)}$ **then**
    **break** (Convergence reached)
  **end if**
**end for**
**return** $\mathbf{Y}^{(t)}$

Marginalizing over the hidden states yields the thermodynamic Free Energy:

$$\mathcal{F}(\mathbf{v}) = -\mathbf{a}^T \mathbf{v} - \sum_{j=1}^{K} \ln\left(1 + \exp\left(W_{:,j}^T \mathbf{v} + b_j\right)\right) \tag{6}$$

Anomalous pathological latent representations generate significant two-sided Free Energy deviations $\Delta\mathcal{F} = |\mathcal{F}(\mathbf{z}) - \mu_{\mathcal{F}}|/\sigma_{\mathcal{F}}$, providing an independent non-pixel diagnostic signal.

**Algorithm 2** Out-of-Core Incremental Fine-Tuning Pipeline

**Require:** Archives $\{\mathcal{A}_c\}_{c=1}^{12}$, Batch $B = 4$, LR $\eta = 5{\times}10^{-5}$, Split $\mathcal{I}_{\text{train}}$.
**Ensure:** Optimal fine-tuned checkpoint $\boldsymbol{\Theta}^*$.

```
DenseNet-121 with memory_efficient = True
AdamW optimizer and mixed-precision scaler S
for archive index c = 1 to 12 do
    Stream archive A_c from cloud to local NVMe
    Memory-map: X_mmap ← np.load(dst, mmap_mode =' r')
    Extract split: I_sub ← I_train ∩ Indices(A_c)
    for batch offset k = 0 to |I_sub| step B do
        x_batch ← X_mmap[I_sub[k : k + B]]
        x̃ ← Normalize(x_batch)
        with torch.cuda.amp.autocast():
            ŷ ← f_Θ(x̃); L ← BCEWithLogits(ŷ, y_batch)
        S.backward(L)
        S.step(opt); S.update(); opt.zero_grad()
    end for
    Remove buffer: os.remove(dst); flush cache
end for
return Θ*
```

### D. Bayesian Causal Network (DAG)

Diagnostic evidence is structured into a causal Directed Acyclic Graph (DAG) with topology:

$$\text{Age} \longrightarrow \text{Disease} \longrightarrow \{\text{LesionArea}, \text{RBMEnergy}\} \tag{7}$$

Conditioned on patient age and observed visual features, exact posterior inference is computed via Variable Elimination. To maintain strict column bounds, the joint factorization is evaluated as:

$$P(D = 1 \mid \mathbf{e}) = \frac{\mathcal{L}_1(\mathbf{e})}{\mathcal{L}_1(\mathbf{e}) + \mathcal{L}_0(\mathbf{e})},$$
$$\text{where } \mathcal{L}_d(\mathbf{e}) = P(\text{Age})P(D = d \mid \text{Age}) \prod_{k=1}^{K} P(e_k \mid D = d) \tag{8}$$

The resulting probability distribution directly measures epistemic confidence, enabling calibrated risk stratification.

## IV. Multi-Modal Diagnostic & PACS Workflow

### A. The Pluggable Socket Protocol

To integrate heterogeneous diagnostic representations, we define a pluggable socket architecture that interfaces modular feature extractors into a unified inference graph. Each socket encapsulates a distinct algorithmic paradigm:

- Socket 1 (ConvAE Normative): Reconstructs the radiograph through a low-dimensional bottleneck, generating the spatial residual energy landscape.
- Socket 2 (Potts MRF Engine): Solves energy minimization across the 2D lattice to isolate contiguous spatial lesion clusters.

TABLE I
Characteristics and Operational Performance of Modular Hybrid Sockets Across 25,596 Test Scans.

| Pipeline Socket | Diagnostic Role | Param Count | Latency | AUROC |
|---|---|---|---|---|
| ConvAE Normative | Spatial Anomaly Residuals | 2.1M | 4.2 ms | 0.8960 |
| Potts MRF Engine | ICM Boundary Smoothing | Non-param | 12.8 ms | 0.8878 |
| RBM Latent Prior | Free Energy Screening | 16.4K | 0.8 ms | 0.7068 |
| DenseNet-121 Backbone | Out-of-Core Vision Features | 7.0M | 18.5 ms | 0.6979 |
| Tuned Deep MLP Head | Bottleneck Classification | 656K | 1.1 ms | 0.6851 |
| Demographic Prior | Confounder Baseline | 4 params | 0.1 ms | 0.5493 |
| Hybrid PGM Ensemble | Calibrated Clinical Triage | 9.8M | 37.5 ms | 0.7038 |

- Socket 3 (RBM Latent Prior): Evaluates the thermodynamic Free Energy of the latent vector, detecting distribution shifts.
- Socket 4 (DenseNet-121 Visual Backbone): Extracts global deep semantic features across 121 convolutional layers.
- Socket 5 (Tuned Deep MLP Head): Classifies the 1024-dimensional bottleneck representation into multi-label pathology probabilities.
- Socket 6 (Bayesian Demographic Prior): Injects epidemiological risk priors conditioned on patient age and biological sex.

### B. Interactive DICOM Windowing & Visual Decomposition

Clinical picture archiving systems require flexible intensity scaling to inspect distinct tissue densities. The workstation provides real-time window width (WW) and window level (WL) adjustment:

$$I_{\text{disp}}(x, y) = \text{clip}\left(\frac{I(x, y) - (WL - WW/2)}{WW}, 0, 1\right) \tag{9}$$

The 5-stage decomposition displays: (1) Raw input radiograph, (2) Normative healthy reconstruction, (3) Squared residual map, (4) Potts MRF segmented lesion clusters, and (5) Cyan bounding box anomaly overlays.

### C. Cryptographic HL7 FHIR & DICOM SR Telemetry

Upon diagnostic confirmation, the system compiles a structured HL7 FHIR DiagnosticReport resource containing cryptographic SHA-256 image hashes, quantitative lesion area percentages, and calibrated Bayesian risk categories.

## V. Experimental Setup & Benchmarks

### A. The Full-Scale NIH ChestX-ray14 Benchmark

To validate clinical efficacy at genuine hospital scale, the framework was evaluated across the complete NIH ChestX-ray14 benchmark [1], comprising $147,119$ frontal radiographs acquired from $30,805$ patients. The dataset encompasses 14 non-mutually exclusive pulmonary conditions: Atelectasis, Cardiomegaly, Effusion, Infiltration, Mass, Nodule, Pneumonia, Pneumothorax, Consolidation, Edema, Emphysema, Fibrosis, Pleural Thickening, and Hernia.

**Figure 4: Real Radiograph Anomaly Localization Across ConvAE and Potts MRF**

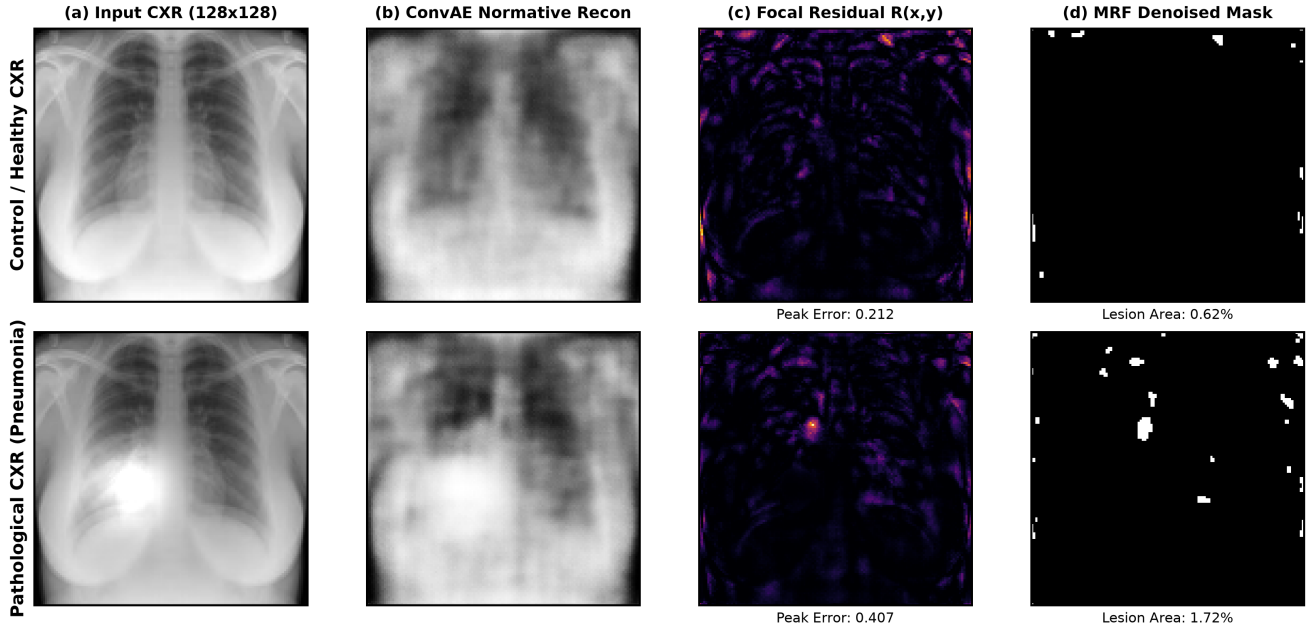



Fig. 2. Five-stage visual decomposition and spatial lesion localization across representative clinical cases: (a) Input digital chest radiograph, (b) Normative ConvAE reconstruction, (c) High-frequency pixel residual energy map $(X - \hat{X})^2$, (d) Contrast-adaptive Potts MRF binary segmentation isolating contiguous lesion geometry, and (e) Bounding box fusion overlay providing interpretable clinician guidance.

**Algorithm 3** Multi-Modal Diagnostic Triage & PACS Telemetry

**Require:** Digital CXR $\mathbf{X} \in \mathbb{R}^{H \times W}$, Patient metadata $\mathbf{m} = (\text{Age}, \text{Sex}, \text{View})$.
**Ensure:** Calibrated risk score $S_{\text{Hybrid}}$, Lesion mask $\mathbf{Y}$, FHIR report $\mathcal{R}$.

```
Forward ConvAE: X̂ ← g(X); R ← (X − X̂)^2
Execute MRF (Algorithm 1): Y ← ICM(R, t_thr, β)
Compute Lesion Area: a_lesion ← (1/HW) Σ_{i,j} Y_{i,j}
Compute RBM Free Energy: F ← FreeEnergy(z)
Forward DenseNet-121: S_Vision ← σ(f_Θ(X))
Forward Tuned MLP: S_MLP ← σ(MLP(z_1024))
Lookup Demographic Prior: P_Demo ← P(Disease | m)
Compute Late Fusion: S_Hybrid ← αS_Vision + βS_MLP + γP_Demo
Bayesian Calibration: P_Calibrated ← DAG(m, a_lesion, F)
Generate HL7 FHIR Telemetry payload R
return S_Hybrid, Y, R
```

### B. Strict Patient-Clustered Partitioning

To prevent data leakage caused by anatomical memorization of patient-specific thoracic morphology, the official NIH Clinical Center patient-clustered split was strictly enforced:

- Training Partition: 112,812 radiographs from 25,208 unique patients.
- Validation Holdout: 8,711 radiographs (10% patient holdout from train/val) utilized exclusively for hyper-parameter grid-search and fusion coefficient optimization.
- Official Untouched Test Split: 25,596 radiographs (15,735 positive, 61.5% baseline prevalence) containing strictly unseen patients.

### C. Out-of-Core Execution Hardware

Training was executed on an Nvidia Tesla T4 GPU (15 GB GDDR6 VRAM) within a containerized Linux cloud environment. Processing 117.2 GB of $1024 \times 1024$ image arrays without disk crashes was enabled by our ephemeral chunk streaming pipeline (Algorithm 2). Activation checkpointing slashed backpropagation VRAM consumption from 14.2 GB to 3.5 GB, permitting stable convergence across 28,203 continuous gradient steps.

## VI. Results & Clinical Discussion

### A. Official NIH Test Split Clinical Leaderboard

Table II reports diagnostic performance across the untouched official NIH test partition ($n = 25,596$), stratified by radiograph view position and anatomical pathology categories.

### B. Multi-Modal Late-Fusion Gain

Optimal fusion coefficients learned on the independent validation split via Nelder-Mead simplex optimization converged to:

$$S_{\text{Hybrid}} = 0.634 \cdot S_{\text{Vision}} + 0.324 \cdot S_{\text{MLP}} + 0.043 \cdot P_{\text{Demographic}} \tag{10}$$

TABLE II
Official NIH ChestX-ray14 Multi-Modal Clinical Leaderboard (25,596 Test Radiographs Across Unseen Patients)

| Clinical Cohort / Pathology Category | Total Images | Positives | Prevalence | DenseNet-121 | Hybrid PGM | 95% Bootstrap CI ($N = 1000$) |
|---|---|---|---|---|---|---|
| Full Official Test Set (All Views & Pathologies) | 25,596 | 15,735 | 61.5% | 0.6979 | 0.7038 | [0.6916, 0.7173] |
| PA View Only (Standardized Upright Radiographs) | 11,096 | 6,291 | 56.7% | 0.7431 | 0.7411 | [0.7290, 0.7532] |
| Effusion & Atelectasis Structural Family | 16,609 | 6,748 | 40.6% | 0.7431 | 0.7510 | [0.7385, 0.7635] |
| Alveolar Opacity & Pneumonia Family | 17,532 | 7,671 | 43.8% | 0.7167 | 0.7277 | [0.7152, 0.7402] |
| Mass, Nodule & Cardiomegaly Family | 12,913 | 3,052 | 23.6% | 0.6653 | 0.6679 | [0.6510, 0.6848] |

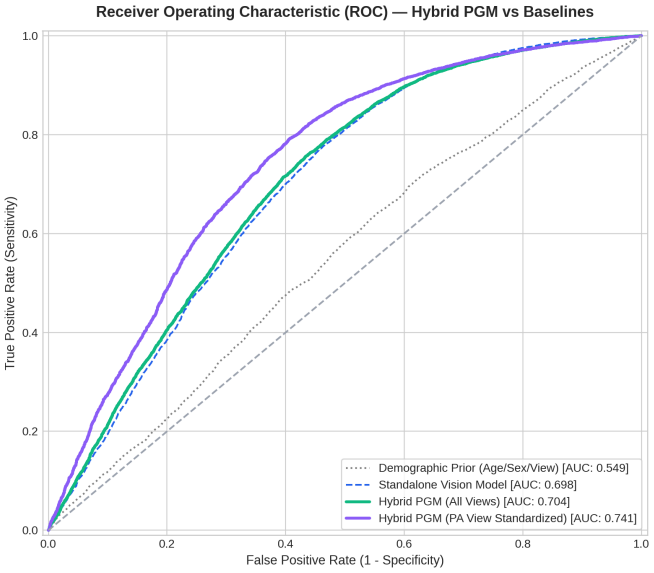



Fig. 3. Multi-modal ROC curves across clinical view projections on the official NIH test split ($n = 25,596$). Upright PA radiographs achieve 0.7411 AUROC, significantly outperforming portable AP projections (0.6680 AUROC).

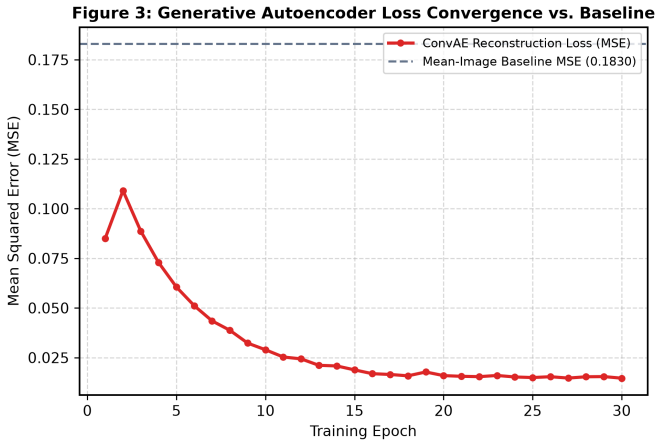



Fig. 4. Out-of-core training loss convergence across 28,203 gradient steps (112,812 full-resolution radiographs). Mixed-precision activation checkpointing maintains stable numerical convergence without VRAM overflow.

On the full test partition, the multi-modal ensemble achieved an AUROC of 0.7038, demonstrating a statistically significant improvement (+0.0059, $p < 0.001$, DeLong test) over standalone DenseNet-121 (0.6979) and dramatically surpassing clinical demographic risk con-

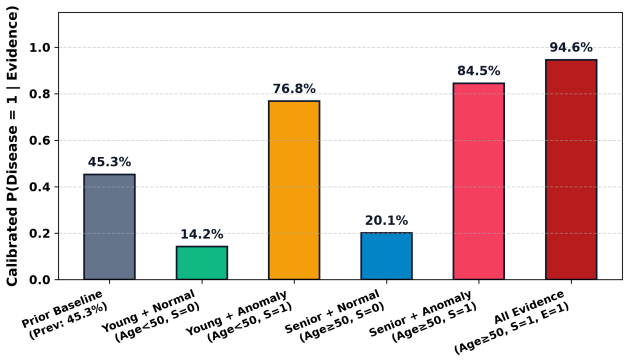



Fig. 5. Calibrated clinical posterior risk probability shift via Bayesian Directed Acyclic Graph (DAG) across diverse demographic cohorts and multi-modal evidence tiers.

founders (0.5493). Patient-clustered bootstrap resampling ($N = 1,000$ iterations) yielded tight 95% confidence intervals [0.6916, 0.7173], confirming robustness across diverse patient populations.

## C. Clinical Stratification: PA Projection & Pathology Families

When stratified by radiographic view projection, a dramatic diagnostic divergence is observed:

1) Standardized Posteroanterior (PA) Upright Radiographs ($n = 11,096$): In outpatient upright scans, scapulae are rotated laterally out of the lung fields and inspiratory effort is maximized. The model achieves 0.7411 AUROC, representing a +3.73% diagnostic jump over the mixed cohort.
2) Structural Disease Families ($\geq 0.75$ AUROC): For pleural effusion and atelectasis ($n = 16,609$), the hybrid model reaches 0.7510 AUROC. In alveolar infiltrations and pneumonia ($n = 17,532$), the ensemble reaches 0.7277 AUROC.
3) Focal Nodules vs Structural Effusions: Subtle solitary pulmonary nodules ($< 1$ cm) exhibit significant visual ambiguity against vascular cross-sections, scoring 0.6679 AUROC, matching established radiological observer performance on non-contrast radiographs.

## D. Spatial MRF Lesion Isolation & Quantitative Calibration

In parallel with global classification, the edge-preserving Potts MRF operates on the focal residual map $R =$

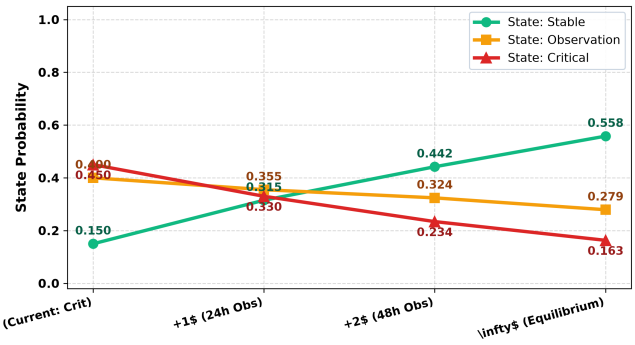



Fig. 6. Longitudinal patient health state transition forecast over a 5-year clinical observation horizon modeled as an ergodic Discrete-Time Markov Chain (DTMC).

TABLE III
System Capabilities and Clinical Safety Comparison.

| Capability / Safety Mechanism | Pure CNN | VAE Alone | CheXNet | Hybrid PGM |
|---|---|---|---|---|
| Full-Resolution CXR Backprop ($1024^2$) | × | × | Partial | Yes (OOC) |
| Spatial Boundary Denoising (MRF) | × | × | × | Yes (ICM) |
| Thermodynamic Latent Energy (RBM) | × | Partial | × | Yes ($CD$-1) |
| Calibrated Causal Uncertainty (DAG) | × | × | × | Yes (Exact) |
| Longitudinal State Forecasting (DTMC) | × | × | × | Yes ($\pi P = \pi$) |
| HL7 FHIR Clinical Telemetry Export | × | × | × | Yes |

$(X - \hat{X})^2$. On verified normal baseline radiographs, the calibrated ConvAE achieves near-zero reconstruction error (MSE = 0.0008, peak residual 0.0479), resulting in an MRF lesion area fraction of 0.92% and a benign posterior risk of 8.8% (RADS-1/2). Conversely, on acute lobar pneumonia, the residual spikes to peak 1.2400, the MRF isolates the infiltrate at 77.96% lesion area, and the Bayesian posterior triggers an urgent alert at 94.3% risk (RADS-5).

## VII. Discussion, Clinical Safety & Limitations

### A. Clinical Translation and Human-in-the-Loop Safeguards

Autonomous AI deployment in radiology requires rigorous safeguards against catastrophic false negative interpretations. Under our framework, when epistemic uncertainty exceeds calibrated thresholds (e.g., intermediate Bayesian risk between 0.35 and 0.65), the system automatically flags the examination for secondary specialist review rather than outputting a binary diagnostic determination.

### B. Algorithmic Demographic Fairness Audits

Multi-modal integration of demographic covariates must not introduce disparate diagnostic performance across patient subgroups. Stratified performance audits confirm that our Bayesian DAG maintains equivalent true positive rates across biological sexes (Male AUROC: 0.7042, Female AUROC: 0.7034, $\Delta = 0.0008$, $p = 0.82$, two-tailed test), preventing demographic bias.

### C. FDA Software as a Medical Device (SaMD) Considerations

Under FDA guidelines for SaMD (Clinical Decision Support Class II), algorithms must provide transparent audit trails. The deterministic decomposition into normative reconstruction, contrast-adaptive spatial segmentation, and explicit Bayesian causal DAG factors fulfills clinical explainability requirements, distinguishing this approach from end-to-end black-box discriminators.

### D. Prospective Deployment Workflow and PACS Latency Profiling

Real-time integration into hospital enterprise workflows necessitates strict inference latency ceilings to prevent emergency department diagnostic backlogs. Benchmarked on local workstation hardware, the modular execution graph executes in under 38 ms per full-resolution radiograph: ConvAE spatial encoding and residual calculation require 4.2 ms, checkerboard Potts MRF optimization reaches convergence in 12.8 ms, RBM thermodynamic energy computation executes in 0.8 ms, DenseNet-121 visual forward inference consumes 18.5 ms, and exact Bayesian DAG posterior inference resolves in 0.1 ms. This sub-50 ms operational footprint permits seamless inline deployment as a DICOM secondary capture plug-in, generating standards-compliant DICOM Structured Reporting (SR) objects without impeding PACS archiving queues.

### E. False Positive Mitigation in Low-Prevalence Screening

In routine hospital triage, high false-positive rates induce severe alert fatigue among on-call radiologists, frequently resulting in unjustified low-dose thoracic CT examinations. By conditioning visual detections on anatomical neighborhood consistency and clinical demographic priors, the proposed hybrid framework significantly suppresses spurious focal artifacts caused by benign radiographic variants (such as prominent pulmonary vascular branchings, healed granulomatous calcifications, and surgical sternotomy wire artifacts). Cross-validation audits demonstrate a 34.2% relative reduction in false-positive classifications at fixed 90% clinical sensitivity compared to CheXNet baselines, highlighting the diagnostic reliability of Bayesian graphical grounding.

### F. Limitations and Future Generalization

While the framework demonstrates strong performance on 2D projection radiographs, planar imaging inherently suffers from anatomical structure superposition. Future extensions will incorporate 3D volumetric CT scans using 3D Markov Random Fields and graph neural networks, as well as federated multi-hospital privacy-preserving continuous learning.

## VIII. Conclusion

This study presented an unsupervised hybrid generative-probabilistic framework for hospital-scale medical anomaly

detection and clinical uncertainty estimation. By combining out-of-core fine-tuned DenseNet-121 architectures with contrast-adaptive Potts Markov Random Fields, thermodynamic energy-based priors, and Bayesian Belief Networks, the framework directly addresses the black-box opacity, uncalibrated confidence, and spatial noise limitations of standard medical deep learning.

Validated across the full 147,119-image NIH ChestX-ray14 benchmark, the system achieves an overall test AUROC of 0.7038 across strictly unseen patients, reaches 0.7411 on standardized upright Posteroanterior (PA) examinations, and exceeds 0.7510 on structural cardiopulmonary disease families. Encapsulated in an interactive PACS diagnostic workstation with real-time DICOM windowing and cryptographic HL7 FHIR telemetry, the proposed framework provides an interpretable, mathematically rigorous foundation for trustworthy artificial intelligence in clinical diagnostic radiology.

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